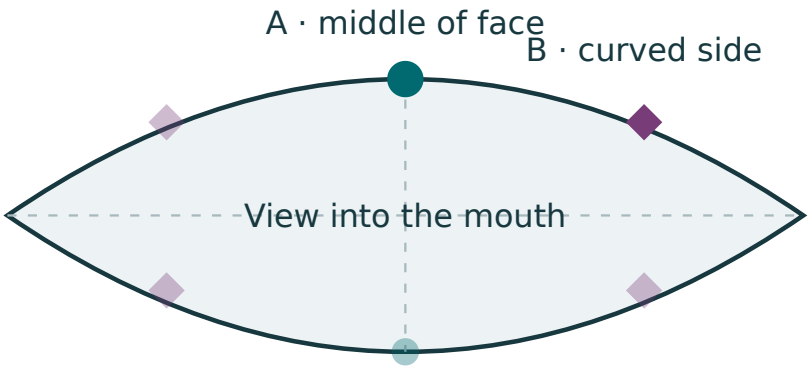
Ten bells. How many notes?

Selected: CsM I, Changsha Museum 1632.9. Shiguzhai, Hunan, China.

> A: 280 Hz (C♯4, +19.4 cents). B: 269 Hz (C4, +50 cents). The selected B is -69.4 cents relative to A. Note names are modern reference labels, with A4 = 439.524 Hz. Moving this reference changes the labels, not the measured frequencies.

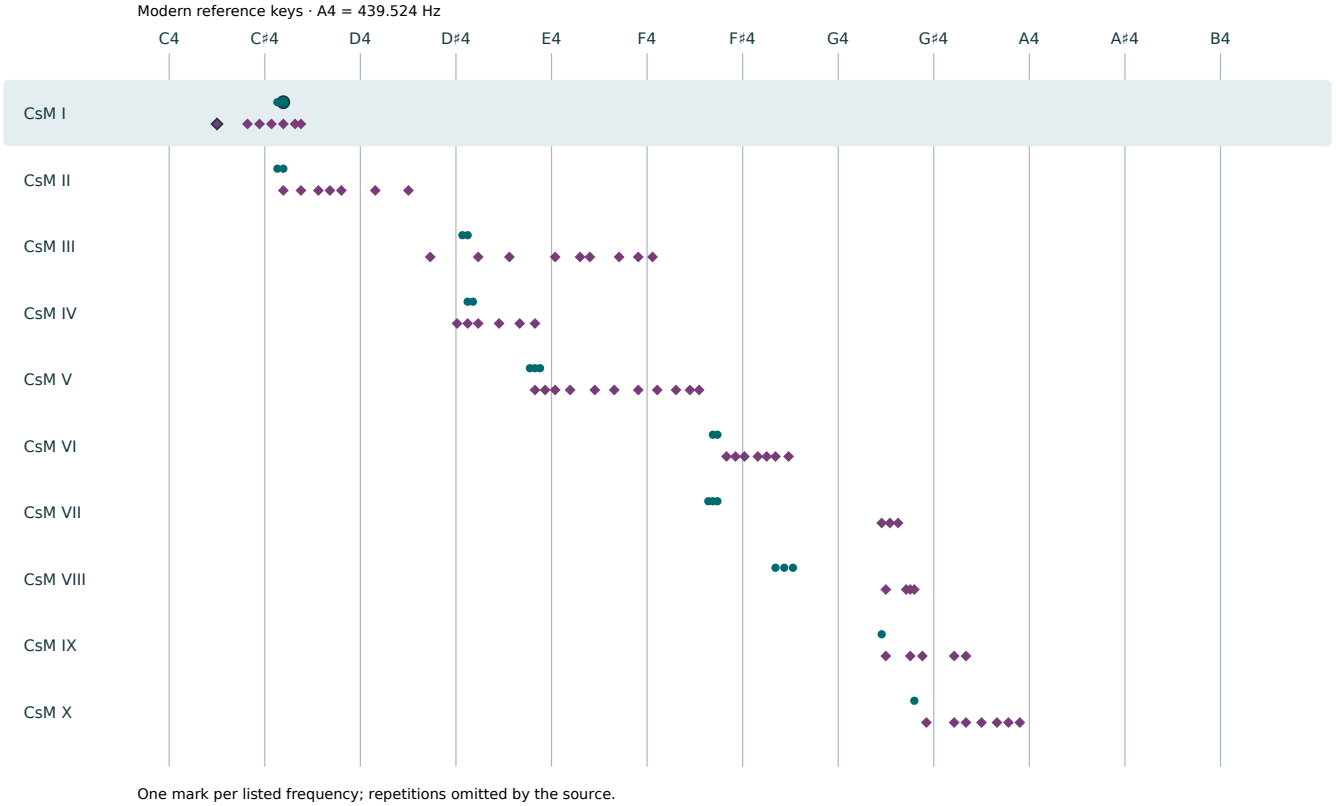


Strike locations only; shape and positions are schematic.

The app plays short sine tones at these frequencies. It does not reproduce the bells' timbre or an ancient performance. The diagram locates strike areas schematically.

Ten bells from one find

Every frequency listed in the source table is retained. 87 plotted entries fall nearest to 10 modern keys in the octave C4–B4. This classification does not prove an ancient equal-tempered scale.



One mark per listed frequency; repetitions omitted by the source.

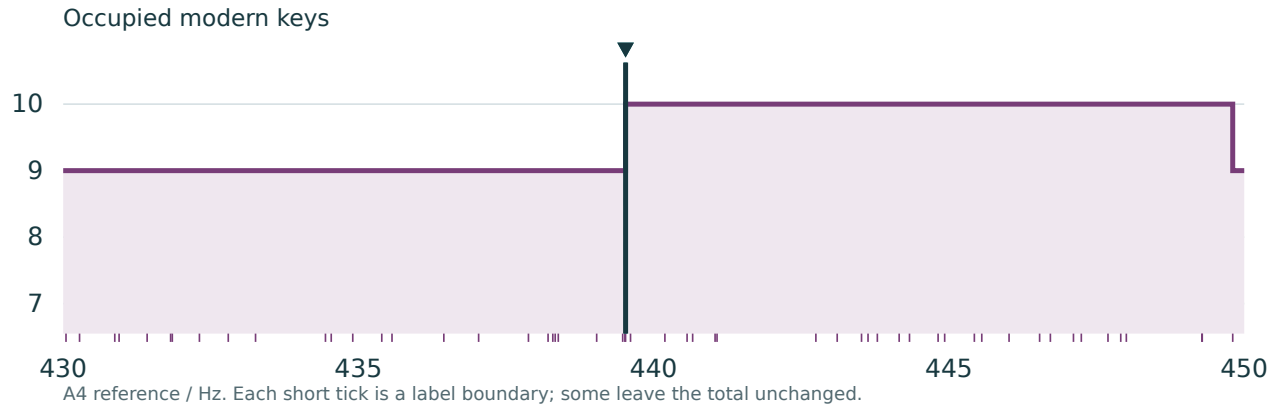
C4	C♯4	D4	D♯4	E4	F4	F♯4	G4	G♯4	A4	A♯4	B4

present	present	present	present	present	present	present	present	present	present	absent	absent

The same measurements, two modern grids

Both sides retain 87 table entries from all ten bells, using every listed frequency. The hertz measurements remain fixed.

Held A4 439.523 Hz: 9 keys. Current A4 439.524 Hz: 10 keys. Changed table entries: 1.



Held A4: 439.523 Hz. Current A4: 439.524 Hz.

CsM entry	Hz	Held	Current
I · B7	269	C♯4	C4

Across the chosen modern 430–450 Hz range, the count varies from 9 to 10. The calculation solves all 54 label boundaries, including those that leave the total unchanged.

Nearest boundary: A4 = 439.523257 Hz for the reported 269 Hz. As A4 rises, CsM I B7 move from C♯4 to C4. The total changes from 9 to 10. The printed decimal is rounded; the calculation uses the unrounded boundary.

This is sensitivity to a modern classification choice. The extra decimal places locate a calculated boundary; they do not increase the precision of the reported integer frequencies. The source supplies no measurement error bars. Table positions such as B7 mean the seventh listed B frequency, not the seventh strike.

Keep the reported frequencies

Values preserve the publisher’s order. First B entries are the clearest or most often occurring; the paper supplies neither frequency occurrence counts nor strike order. The first A entry has no equivalent explicit priority claim.

Bell / accession	A / Hz	B / Hz	In chosen group?
CsM I 1632.9	280, 279	283, 282, 280, 278, 276, 274, 269	Yes
CsM II 1632.5	280, 279	296, 302, 290, 288, 286, 283, 280	Yes
CsM III 1633	313, 312	350, 347, 343, 337, 335, 330, 321, 315, 306	Yes

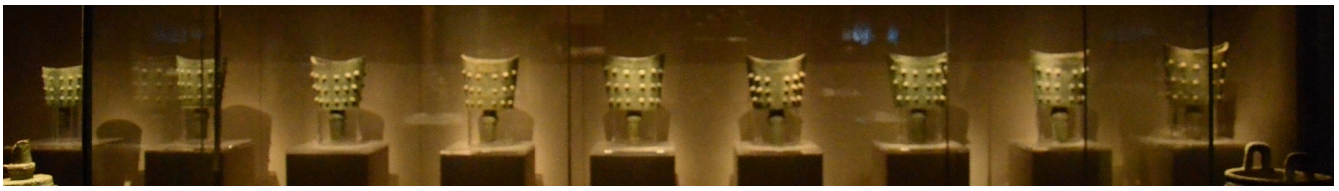
Bell / accession	A / Hz	B / Hz	In chosen group?
CsM IV 1632.8	313, 314	326, 323, 319, 315, 313, 311	Yes
CsM V 1632.7	325, 326, 327	355, 360, 358, 351, 347, 342, 338, 333, 330, 328, 326	Yes
CsM VI 1632.4	364, 363	377, 380, 375, 373, 370, 368, 366	Yes
CsM VII 1632.3	362, 363, 364	406, 404, 402	Yes
CsM VIII 1632.2	379, 381, 377	409, 410, 408, 403	Yes
CsM IX 1632.6	402	423, 420, 412, 409, 403	Yes
CsM X 1632.1	410	437, 434, 431, 427, 423, 420, 413	Yes

Read the evidence in context

The paper reports measurements made in August 2022 at 22 °C, using a wooden mallet and laptop recording. These are observations of surviving objects, not measurements taken in antiquity. Its authors argue that the bells could form a musical set; grouping, surviving condition and interpretation remain part of that argument.

The ten objects were found together in June 1993. The study dates their manufacture to the twelfth century BCE. Five other bells found in 1959 and two found in August 1993 make seventeen from the hill; this comparison includes only the ten with the June-find identities and full frequency lists.

Objects before categories



Nine bells with knob decoration. Li et al. (2024), figure 5. Photograph by Kin Sum Li, CC BY 4.0.



The differently decorated CsM III, Changsha Museum 1633. Li et al., figure 2. Photograph by Kin Sum Li, CC BY 4.0. The complete photograph is retained.

Names to carry with you

鐃 (**náo**): the mouth-up bell type studied here. 鐘 (**zhōng**): bell. 編鐘 (**biānzhōng**): an ordered set of bells. These are modern Mandarin readings, not reconstructed Bronze Age speech. See the app’s linked dictionaries and museum entry for the terms.

The published table

Li et al. (2024), Table 1, page 6. This page includes two additional bells outside the ten-bell comparison. Rotated for reading; text and content unchanged.

Table 1 Pitch test of some bells found in the Ningxiang county

Corresponding author's new no. and appellation	Ningxiang-Shiguzhai-elephant1	Ningxiang-Shiguzhai-Feng2	CsM I	II	III	IV	V	VI	VII	VIII	IX	X
A-tone (Hz; one to two of them will be turned into musical notes)	G#3–45	A#3–31 (229; 228)	C#4 + 18 (280; 279)	C#4 + 18 (280; 279)	D#4 + 10 (313; 312)	D#4 + 10 (313; 314)	E4-24 (325; 326, 327)	F#4–28 (364; 363)	F#4–38 (362; 363, 364)	F#4 + 42 (379; 381, 377)	G4 + 44 (402)	G#4–22 (410)
B-tone (the number before the semi-colon is the frequency that is the clearest or most often occurred)	A#3–23 [not tested by the corresponding author, so no Hz is given.]	B3-49 (240; 237, 235, 233, 230, 229, 228, 227, 226)	C#4 + 36 C4 + 48 (283; 282, 280, 278, 276, 274, 269)	D4 + 14 C#4 + 18 (296; 302, 290, 288, 286, 283, 280)	F4 + 4 D#4–29 (350; 347, 343, 337, 335, 330, 321, 315, 306)	E4-19 D#4–1 (326; 323, 319, 315, 313, 311)	F4 + 28 E4-19 (355; 360, 358, 351, 347, 342, 338, 333, 330, 328, 326)	F#4 + 32 F#4–19 (377; 380, 375, 373, 370, 368, 366)	G#4–39 G4 + 44 (406; 404, 402)	G#4–26 G4 + 48 (409; 410, 408, 403)	G#4 + 32 G4 + 48 (423; 420, 412, 409, 403)	A4-12 G#4–10 (437; 434, 431, 427, 423, 420, 413)
Accession no	HnM 39202	Wenguansuo 0667	CsM 1632.9	1632.5	1633	1632.8	1632.7	1632.4	1632.3	1632.2	1632.6	1632.1
Side length (*xianchang*)	44.5	?	34	31	31.5	31	31	29.5	28	28.3	29	23.3
Spine length (*zhongchang*)	40	?	29.6	27.5	28	27.5	27	26.5	25.5	24.8	26	21
Girth of lip-mid-hip (LMH)	?	?	99–90-79	90.2–84-72	96.5–89.5–79.5	90–83-72.5	90.5–83.5–71	84.5–79-69.5	84–75.4–63.3	80.5–74.5–64.2	86.5–80.5–70.2	66.5–61.5–54.3
Xianjian, gujian (cavity)	?	?	33.5, 19	32, 21	32, 23	30, 21	32, 20	28.5, 20	28.5, 18.7	27, 19	29.5, 20	23, 15
Wuxiu, wuguang (cavity)	?	?	24, 19	24, 16	25.5, 18	24, 16	24, 15	23, 15.5	23, 13.5	21.5, 14.5	23, 16.5	18, 13
Overall height	70+	80	53.5	48.5	54	51.6	50	48	45.7	47	49.5	36.5

Most of the recorded data was measured and tested by the corresponding author in August 2022 in the exhibition gallery of the Changsha Museum

Temperature of the testing location: 22 °C

All measurements are presented in centimeters, unless indicated otherwise

In the LMH section, the term "lip" refers to the girth of the mouth of the bell, "mid" indicates the girth of the section beneath the top two rows of knobs, and "hip" denotes the girth of the lowest section of the bell's sounding body

A4 = 440 Hz

Overtones of any resonance are not considered the intended tones of a bell; only the fundamental tone is documented

For testing [3, 6–8, 28–31], each bell was placed on the floor with its shank at the base. The A-tone strike position on one side of the bell was struck three times, followed by three strikes on the other side. The corresponding author then tested the four B-tone strike positions, striking each position three times. The sound generated by the bells was recorded and analyzed using the internal recording device of a Samsung Galaxy Book Pro laptop and the audial software "Audacity" [32]

Kin Sum Li and colleagues, A 'set' of ancient bronze bells excavated in Changsha, Hunan Province, China, *Heritage Science* 12, 264 (2024). CC BY 4.0; the article applies CC0 to its data unless indicated otherwise.

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